

23.6 When resampling fails

The consistency theorems of this chapter assume exchangeable resampled quantities, a smooth statistic, a fixed number of parameters, a finite error variance, no dominant case, independence, and a sample large enough for limits to be relevant. When any of these fails, the bootstrap can fail silently, producing a distribution that looks reasonable and is wrong.

23.6.1 A statistic that is not smooth: the maximum

The cleanest failure is not specific to regression. Let $Y_1, \dots, Y_n$ be independent and uniform on $[0, \theta]$, and estimate θ by the sample maximum $Y_{(n)}$.

Proposition 23.6.1. The bootstrap of a maximum

- (a) $n(\theta - Y_{(n)})/\theta$ converges in distribution to the standard exponential law.
- (b) For every sample with distinct values, the case bootstrap gives $\Pr_*(Y_{(n)}^* = Y_{(n)}) = 1 - (1 - 1/n)^n \rightarrow 1 - e^{-1}$.

So the bootstrap distribution of $n(Y_{(n)} - Y_{(n)}^*)/Y_{(n)}$ has an atom at zero of mass at least $1 - e^{-1}$ for every sample, and it cannot approach the continuous exponential law.

Proof.

- (a) For $0 \leq t \leq n$, $\Pr(n(\theta - Y_{(n)})/\theta > t) = \Pr(\text{all } Y_i < \theta(1 - t/n)) = (1 - t/n)^n \rightarrow e^{-t}$.
- (b) $Y_{(n)}^* = Y_{(n)}$ unless none of the n draws picks the largest observation, which happens with probability $(1 - 1/n)^n$. And $(1 - 1/n)^n = \exp(n \log(1 - 1/n)) \rightarrow e^{-1}$, from below, since $n \log(1 - 1/n) < -1$.

■

With $n = 50$ the atom is $1 - 0.98^{50} = 0.636$, and a simulation with 20000 resamples finds 0.639. The bootstrap needs the statistic to depend smoothly on the empirical distribution, and extremes do not. The m -out-of- n bootstrap, with resamples of size m and $m/n \rightarrow 0$, repairs many such failures at a price in efficiency (Bickel, Götze and van Zwet 1997). In regression the analogue is any statistic driven by the most extreme residual, such as the largest studentized residual of the outlier test (Theorem 20.3.3).

```
import numpy as np
rng = np.random.default_rng(2310)
n = 50
u = rng.uniform(size=n)
B = 20_000
boot_max = u[rng.integers(0, n, size=(B, n))].max(axis=1)
atom = np.mean(boot_max == u.max())
print(f"bootstrap P*(max* = max) = {atom:.3f}; 1 - (1 - 1/n)^n = {1 - (1 - 1 / n) ** n:.3f}")
```

[Run this code online](#)

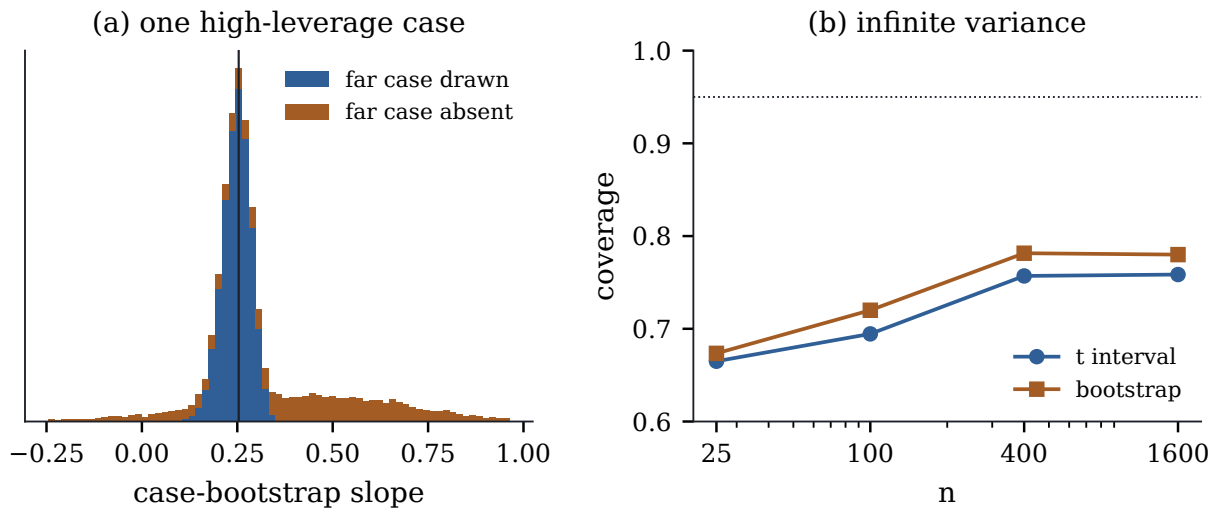


Figure 23.6.1: (a) Case-bootstrap slopes when one of 15 cases has leverage 0.902, split by whether the resample contains it. (b) Coverage of nominal 95% intervals for a Pareto mean with infinite variance.

23.6.2 Heavy tails

Without a finite error variance the sample mean, and any least squares coefficient, is not asymptotically normal. Athreya (1987) showed that the bootstrap of the mean then converges not to the correct stable limit but to a *random* limit that differs from sample to sample: resampling cannot learn a tail that shows up only through a few enormous values.

Figure 23.6.1(b) shows the consequence for coverage. The data are samples from a Pareto law with tail index 1.5, which has a finite mean and an infinite variance, and the target is the mean. The nominal 95% t interval covers 0.665 of the time at $n = 25$ and 0.758 at $n = 1600$; the bootstrap percentile interval covers 0.673 and 0.780. Neither improves between $n = 400$ and $n = 1600$. The same holds for regression coefficients, which is one reason to prefer estimators less sensitive to extreme errors, such as the least absolute deviations fit of Chapter 45 (Definition 45.2.1), to resampling least squares.

23.6.3 One dominant case

When one case has leverage near one, Eq. 23.1.1 fails and the case bootstrap splits into resamples that contain the case and resamples, about a third of them, that do not.

Example. A slope determined by one case

Take 15 cases, fourteen with x spread over $[0, 1]$ and one at $x = 4$, with leverage 0.902, and a response that puts the far case a little below the line of the others. The least squares slope is 0.254. Without the far case it would be 0.437. In 20000 case-bootstrap resamples, a fraction 0.356 omit the far case, close to $(14/15)^{15} = 0.355$. Figure 23.6.1(a) shows the result. The resamples that contain the far case give slopes tightly concentrated near the full-data value, with standard deviation 0.037. Those that omit it give slopes spread around the fourteen-case value, with standard deviation 0.274. The bootstrap distribution mixes two answers to two questions, and neither its spread nor its percentiles describe the uncertainty of $\hat{\beta}_1$.

```
import numpy as np
```

```

rng = np.random.default_rng(2313)
n_l = 15
x_l = np.append(np.linspace(0, 1, n_l - 1), 4.0)    # one case far from the rest
X_l = np.column_stack([np.ones(n_l), x_l])
h_l = np.sum((X_l @ np.linalg.inv(X_l.T @ X_l)) * X_l, axis=1)
y_l = 1 + 0.5 * x_l + 0.3 * rng.normal(size=n_l)
y_l[-1] -= 1.0                                     # and not quite on the line of
                                                    the others
idx = rng.integers(0, n_l, size=(20_000, n_l))
xs, ys = x_l[idx], y_l[idx]
xc = xs - xs.mean(axis=1, keepdims=True)
b_case = np.sum(xc * ys, axis=1) / np.sum(xc ** 2, axis=1)
absent = ~np.any(idx == n_l - 1, axis=1)
print(f"leverage of the far case {h_l[-1]:.3f}; resamples without it: {absent.mean():.3f}")

```

[Run this code online](#)

The residual and wild bootstraps keep the far case, but inherit the problem (Example (The wild bootstrap for Engel's data)): its residual has variance $\sigma^2(1 - h_{ii})$, nearly zero, so the data say almost nothing about its error.

23.6.4 Many parameters

With p proportional to n , the plain residual bootstrap underestimates variances by the non-vanishing factor $(n - p)/n$ of Proposition 23.1.1(b). The case bootstrap errs the other way: a resample has only about $0.632n$ distinct cases (Exercise (A2)), and when p/n is not small its design is much worse conditioned than the original.

A simulation with $n = 60$, $p = 30$, a Gaussian design and normal errors shows both effects. The median ratio of bootstrap variance to true variance for one coefficient, over 200 data sets, is 0.47 for the plain residual bootstrap, 0.97 after rescaling the residuals by $\sqrt{n/(n - p)}$, and 3.38 for the case bootstrap. El Karoui and Purdom (2018) study these failures, and Mammen (1993) gives conditions under which the residual and wild bootstraps remain valid as p grows. [Chapter 28](#) takes up regression with more parameters than cases.

```

import numpy as np

def variance_ratios(n, p, reps, B, rng):
    """Bootstrap variance of the first coefficient divided by its true variance (
    errors  $N(0, 1)$ )."""
    X = rng.normal(size=(n, p))                    # fixed design
    XtX_inv = np.linalg.inv(X.T @ X)
    true_var = XtX_inv[0, 0]
    H = X @ XtX_inv @ X.T
    out = {"residual": [], "rescaled": [], "case": []}
    for _ in range(reps):
        y = rng.normal(size=n)                    # beta = 0
        e = y - H @ y
        fit = H @ y
        for name, r in [("residual", e), ("rescaled", e * np.sqrt(n / (n - p)))]:
            Ys = fit + r[rng.integers(0, n, size=(B, n))]
            out[name].append(np.var(Ys @ (X @ XtX_inv)[: , 0]) / true_var)
    bs = []
    for _ in range(B):
        i = rng.integers(0, n, size=n)

```

```

        bs.append(np.linalg.lstsq(X[i], y[i], rcond=None)[0][0])
    out["case"].append(np.var(bs) / true_var)
    return {k: float(np.median(v)) for k, v in out.items()}

rng = np.random.default_rng(2314)
print(variance_ratios(60, 30, reps=20, B=100, rng=rng))

```

[Run this code online](#)

23.6.5 Dependence

Resampling independently destroys serial correlation. With positively autocorrelated errors and a smooth regressor, the true variance of a slope far exceeds the independent-errors formula (Theorem 19.2.2), which every bootstrap of this chapter reproduces. Take a straight line in time with $n = 100$ equally spaced points and AR(1) errors with parameter 0.7. Over 2000 series the slope has standard deviation 1.130, while the residual bootstrap's average standard error is 0.449, a ratio of 0.40.

The **moving-block bootstrap** (Künsch 1989) resamples blocks of consecutive residuals and so keeps the dependence within blocks. With blocks of length 10 its average standard error is 0.818, a ratio of 0.72: better, still short, because dependence between blocks is lost (Lahiri 2003). When the dependence has a known form it is better to model it, as in Theorem 21.6.1, and resample the estimated innovations.

```

import numpy as np

def dependence_study(n, phi, reps, B, block, rng):
    """True sd of the slope, and the average iid and moving-block residual-bootstrap
    standard errors."""
    x = np.linspace(0, 1, n)
    X = np.column_stack([np.ones(n), x])
    A = np.linalg.solve(X.T @ X, X.T)
    slopes, se_iid, se_block = [], [], []
    starts_max = n - block + 1
    for _ in range(reps):
        eps = np.empty(n)
        eps[0] = rng.normal() / np.sqrt(1 - phi ** 2)
        for i in range(1, n):
            eps[i] = phi * eps[i - 1] + rng.normal()    # AR(1) errors
        y = 1 + 2 * x + eps
        b = A @ y
        e = y - X @ b
        slopes.append(b[1])
        Ys = X @ b + e[rng.integers(0, n, size=(B, n))]
        se_iid.append(np.std(Ys @ A[1]))
        starts = rng.integers(0, starts_max, size=(B, n // block))
        idx = (starts[:, :, None] + np.arange(block)).reshape(B, -1)
        Ys = X @ b + e[idx]
        se_block.append(np.std(Ys @ A[1]))
    return np.std(slopes), np.mean(se_iid), np.mean(se_block)

rng = np.random.default_rng(2316)
print(dependence_study(100, 0.7, reps=100, B=200, block=10, rng=rng))

```

[Run this code online](#)

23.6.6 Small samples

A bootstrap distribution computed from eight residuals is an estimate from eight numbers. Take a straight line with $n = 8$ equally spaced points and normal errors, the most favourable case. In 4000 simulated data sets, the classical t interval covers 0.953, as it must. The residual bootstrap percentile interval covers only 0.857. The reason is Proposition 23.1.1(b) combined with Proposition 23.3.3(b). The percentile interval is roughly $\hat{\beta}_1 \pm 1.96\sqrt{(n-2)/n} s\|\mathbf{a}\|$, so its coverage is about

$$\Pr(|t_6| \leq 1.96\sqrt{6/8}) = \Pr(|t_6| \leq 1.697) = 0.859.$$

The studentized residual bootstrap covers 0.952. In small samples, studentize, and compare with the classical interval.

```
import numpy as np
from scipy import stats

def small_n_coverage(n, reps, B, rng):
    """Coverage of 95% intervals for a slope with normal errors: t, residual
    percentile, bootstrap-t."""
    x = np.linspace(0, 1, n)
    X = np.column_stack([np.ones(n), x])
    XtX_inv = np.linalg.inv(X.T @ X)
    A = XtX_inv @ X.T
    c = np.sqrt(XtX_inv[1, 1])
    hits = {"t": 0, "percentile": 0, "studentized": 0}
    for _ in range(reps):
        y = rng.normal(size=n)           # true slope 0
        b = A[1] @ y
        e = y - X @ (A @ y)
        s = np.sqrt(e @ e / (n - 2))
        hits["t"] += abs(b) <= stats.t.ppf(0.975, n - 2) * s * c
        Ys = X @ (A @ y) + e[rng.integers(0, n, size=(B, n))]
        bs = Ys @ A[1]
        ss = np.sqrt(np.sum((Ys - (Ys @ A.T) @ X.T) ** 2, axis=1) / (n - 2))
        lo, hi = np.quantile(bs, [0.025, 0.975])
        hits["percentile"] += lo <= 0 <= hi
        tl, th = np.quantile((bs - b) / (ss * c), [0.025, 0.975])
        hits["studentized"] += b - th * s * c <= 0 <= b - tl * s * c
    return {k: v / reps for k, v in hits.items()}

rng = np.random.default_rng(2318)
print(small_n_coverage(8, reps=300, B=499, rng=rng))
```

[Run this code online](#)

Idea. Before trusting a bootstrap, check each assumption listed at the start of this section, and studentize the root. A permutation test needs only exchangeability under the null hypothesis, and is exact or not at all.

23.6.7 Exercises

23.6.7.1 A. Check your understanding

Exercise. A1

Compute $1 - (1 - 1/n)^n$ for $n = 10$ and $n = 1000$. Does the failure of [Proposition 23.6.1](#) become less severe with more data?

Solution. The values are 0.651 and 0.632. The atom decreases slightly toward its limit $1 - e^{-1}$ but never vanishes, so the failure does not go away.

Exercise. A2

In the moving-block bootstrap with blocks of length ℓ , explain why the bootstrap variance of a sample mean tends to be too small when ℓ is much shorter than the range of dependence, and why a very long block makes the estimate noisy.

23.6.7.2 B. Practice**Exercise. B1**

In [Example \(A slope determined by one case\)](#), show that the probability that the far case appears at least twice in a resample is $1 - (1 - 1/n)^n - (1 - 1/n)^{n-1}$, and evaluate it for $n = 15$. What does a resample with the far case drawn three times estimate?

Solution. The number of times the case is drawn is binomial with n trials and probability $1/n$. It is zero with probability $(1 - 1/n)^n$ and one with probability $n(1/n)(1 - 1/n)^{n-1} = (1 - 1/n)^{n-1}$, which gives the formula. For $n = 15$ this is $1 - 0.3553 - 0.3806 = 0.2641$. A resample with the case drawn three times fits a line in which that case has triple weight, a weighted least squares estimate that pulls even harder toward the far case, and not a draw from anything like the sampling distribution of $\hat{\beta}_1$.

23.6.7.3 C. Going deeper**Exercise. C1**

For the uniform maximum of [Proposition 23.6.1](#), show that the m -out-of- n bootstrap with $m \rightarrow \infty$ and $m/n \rightarrow 0$ gives $\Pr_*(Y_{(m)}^* = Y_{(n)}) = 1 - (1 - 1/n)^m \rightarrow 0$, and that the bootstrap distribution of $m(Y_{(n)} - Y_{(m)}^*)/Y_{(n)}$ approaches the standard exponential law.